Experiment No. 5
Experiments on Packet Capture Tool Wireshark

Aim

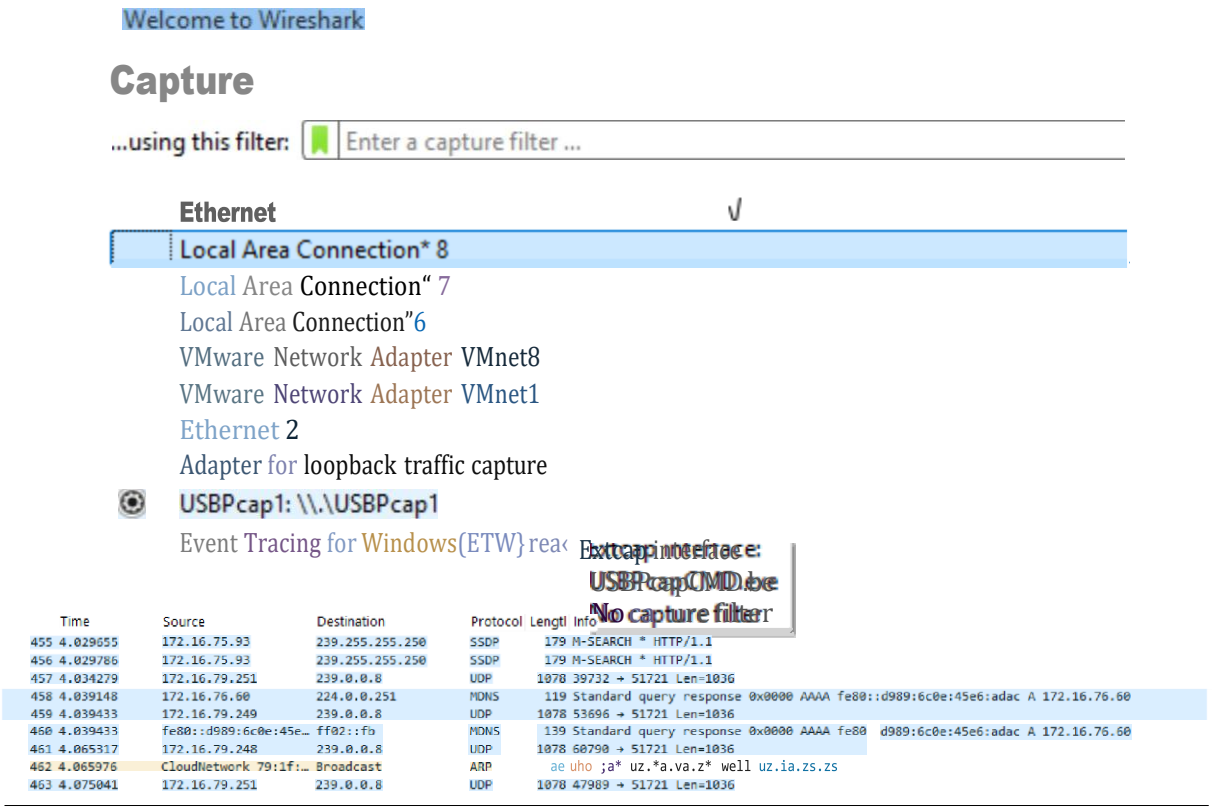
To study the features of Wireshark and analyze packet encapsulation at various layers of the protocol stack.

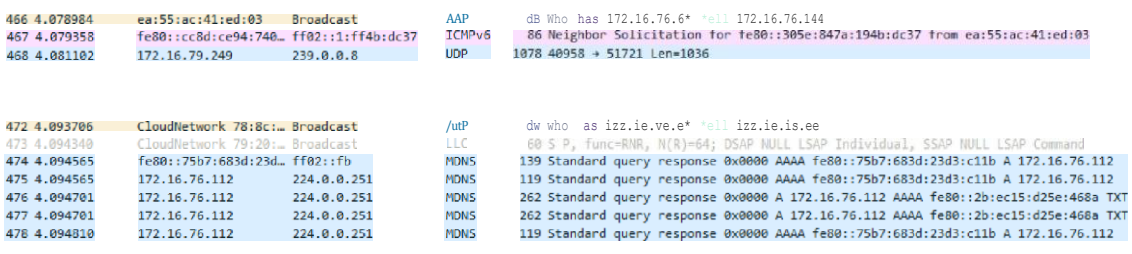
Procedure

1. Download and install Wireshark on the computer. Open the application and become familiar with its interface, packet list pane, packet details pane, and packet bytes pane.
2. Select the active network interface, such as Wi-Fi or Ethernet, and start capturing packets.
3. Generate network traffic by opening a website or executing commands such as ping in the command prompt.
4. Observe the captured packets displayed in the packet list. Note the source address, destination address, protocol, length, and information fields.
5. Apply display filters to identify specific types of packets. For example:
    - icmp – displays ICMP packets.
    - tcp – displays TCP packets.
    - udp – displays UDP packets.
    - dns – displays DNS packets.
    - http – displays HTTP packets when present in the capture.
6. Select an individual packet and examine its details in the packet details pane. Observe the protocol layers and the information contained in each layer.
7. Study encapsulation at different layers of the protocol stack:
    - **Data Link Layer:** Examine the Ethernet header and source and destination MAC addresses.
    - **Network Layer:** Examine the IP header and source and destination IP addresses.

- **Transport Layer:** Examine TCP or UDP headers, including source and destination port numbers.
- **Application Layer:** Observe protocol information such as DNS queries or HTTP requests when available.

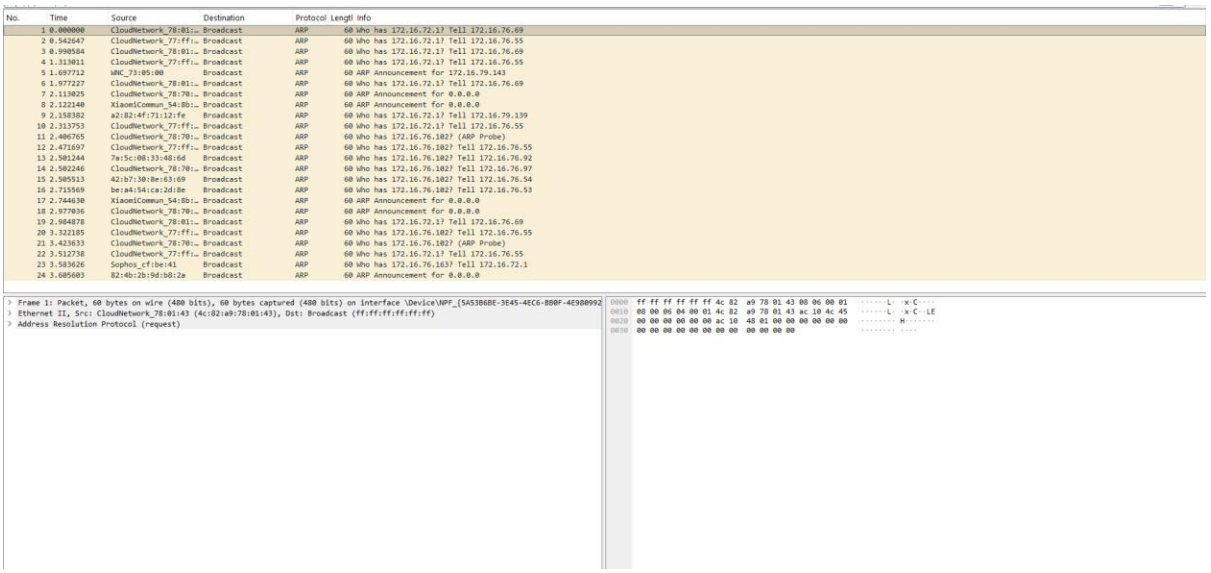
8. Compare packets belonging to different protocols and observe how each layer adds its own header information during encapsulation.
9. Stop the capture and save the captured packets for further analysis.





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Result

Wireshark was used to capture and analyze network packets successfully. Its filtering and packet inspection features were studied, and packet encapsulation at different layers of the protocol stack was observed.